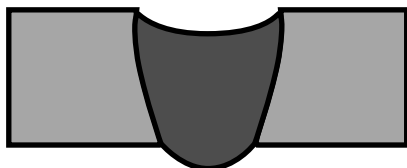


Wire Weld Penetration (Workpiece Heat Control)

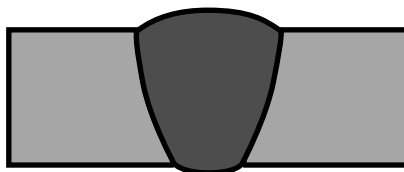
EXCESS PENETRATION OR BURN-THROUGH

Weld droops on top and underneath, or falls through entirely, making a hole.



PROPER PENETRATION

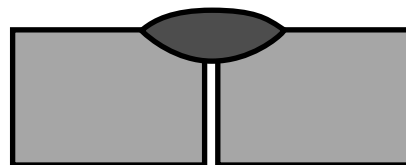
Weld is visible underneath and bulges slightly on top.



PROFILE VIEWS

INADEQUATE PENETRATION

Weld does not penetrate the joint fully, just on the surface.



POSSIBLE CAUSES AND SOLUTIONS

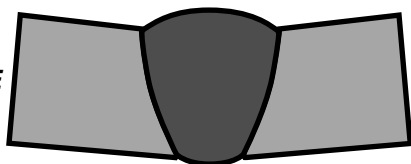
1. **Workpiece overheating:**
Reduce wire feed speed.
Decrease weld current.
2. **Travel speed too slow:**
Increase travel speed and ensure that travel speed is kept steady.
3. **Excessive material at weld:**
Reduce wire feed speed.

POSSIBLE CAUSES AND SOLUTIONS

4. **Incorrect welding technique:**
Maintain 1/2" or less CTWD.
Keep arc on leading edge of weld puddle.
Hold MIG Gun at proper angles.
5. **Insufficient weld heat:**
Reduce travel speed.
Increase weld current.
6. **Workpieces too thick/close:**
Bevel thick workpieces, allow slight gap, and weld in several passes.
7. **Insufficient weld material:**
Increase wire feed speed.

Bend at Joint

PROFILE VIEW



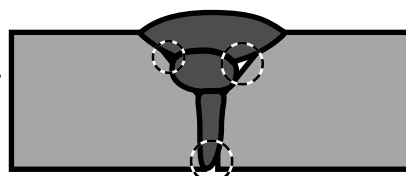
POSSIBLE CAUSES AND SOLUTIONS

1. **Improper clamping:**
Clamp workpieces securely.
Make tack welds to hold workpieces.
2. **Excessive heat:**
Weld a small portion and allow to cool before proceeding.
Increase travel speed.
Reduce wire feed speed.

Weld Not Adhering Properly

Gaps present between weld and previous bead or between weld and workpiece. See areas below.

PROFILE VIEW



POSSIBLE CAUSES AND SOLUTIONS

1. **Incorrect welding technique:**
Place stringer bead at correct place in joint.
Adjust workpiece position or weld angle to permit proper welding to bottom of piece.
Pause briefly at sides during weave bead.
Keep arc on leading edge of weld puddle.
Hold MIG Gun at proper angles.
2. **Insufficient weld heat:**
Increase current.
Increase wire feed speed.
3. **Dirty workpiece:**
Clean workpiece down to bare metal.
4. **Insufficient weld material:**
Increase wire feed speed.
5. **Workpiece gap too narrow:**
Widen groove or increase bevel.

Coat of Slag Over Weld

TOP VIEW



PARTIALLY CHIPPED AWAY TO SHOW WELD

Slag is a necessary part of a flux-cored wire weld. It shields the weld from impurities. Clean off the slag with a Chipping Hammer and Wire Brush after welding.

Gas-shielded MIG welds are protected by the shielding gas and do not need slag to protect them.